A SHORT STUDY OF

Generative AI

GANs, Diffusion Models,
and Transformers

A simple introduction to three important technologies shaping modern artificial intelligence.

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1. Introduction

Artificial intelligence has grown beyond simply recognising patterns or making predictions. Today, it can also create new content, including images, text, music, and videos. This area is known as *Generative Artificial Intelligence*.

Three important technologies behind this progress are Generative Adversarial Networks (GANs), Diffusion Models, and Transformers. Each works differently, but all have contributed to the development of modern AI systems.

2. Generative Adversarial Networks (GANs)

2.1 What is a GAN?

A Generative Adversarial Network, commonly called a GAN, is a model that learns to create new data similar to examples it has seen during training.

It consists of two neural networks that work against each other:

- **Generator:** Creates artificial samples, such as images of faces that do not exist.
- **Discriminator:** Examines samples and tries to determine whether they are real or generated.

The idea is similar to a student and an examiner. The generator keeps improving its work, while the discriminator becomes better at identifying mistakes. Over time, this competition can help the generator produce increasingly realistic results.

2.2 How Does a GAN Work?

The generator begins with a random input, usually called a latent vector, and transforms it into a sample. The discriminator then evaluates that sample.

The training objective is commonly written as

$$\min_G \max_D V(D, G)$$

where G represents the generator and D represents the discriminator.

In simple terms, the generator tries to fool the discriminator, while the discriminator tries not to be fooled. Both networks improve through repeated training.

2.3 Where Are GANs Used?

Some common applications of GANs include:

- Creating realistic artificial images.
- Improving the resolution of photographs.
- Restoring damaged or incomplete images.
- Changing the style of an image.
- Generating synthetic data for research.

2.4 Advantages and Limitations

One of the biggest advantages of GANs is that they can generate sharp and realistic images very quickly once training is complete.

However, training them can be difficult. Sometimes the generator produces similar outputs repeatedly, a problem known as *mode collapse*. Achieving a good balance between the two networks is also an important challenge.

3. Diffusion Models

3.1 What is a Diffusion Model?

Diffusion models take a different approach to generating content. Instead of making a sample in one step, they learn how to gradually remove noise from data.

Imagine taking a clear photograph and slowly covering it with random visual noise until the original image is almost impossible to recognise. A diffusion model learns the reverse process: starting with noise and gradually turning it into a meaningful image.

3.2 How Do Diffusion Models Work?

The process has two main stages:

- **Forward process:** Random noise is gradually added to the original data.
- **Reverse process:** A trained neural network learns to remove the noise step by step.

A simplified mathematical expression for the noisy sample is

$$x_t = \sqrt{\bar{\alpha}_t} x_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon$$

Here, x_0 is the original sample, x_t is the noisy sample at time step t, and ϵ represents random noise.

During generation, the model begins with random noise and repeatedly applies its learned denoising process. The result can be a completely new image or another type of data.

3.3 Where Are Diffusion Models Used?

Diffusion models have become popular in many areas:

- Generating images from text descriptions.
- Editing and restoring photographs.
- Creating video and animation.
- Generating music and other audio.
- Supporting scientific and research applications.

3.4 Advantages and Limitations

Diffusion models can generate detailed, diverse, and high-quality results. They also provide useful ways to edit or condition the generated content.

Their main drawback is speed. Traditional diffusion models may require many denoising steps, which can make generation computationally expensive.

4. Transformers

4.1 What is a Transformer?

A Transformer is a neural network architecture that has changed the way computers process language and other kinds of data.

It became especially important in language models, translation systems, chatbots, and modern AI assistants. Transformers can also be used in image generation, image recognition, and multimodal applications.

Their main idea is called *attention*. It allows the model to focus on different parts of the input depending on what is important for the current task.

4.2 How Do Transformers Work?

When processing text, a Transformer first breaks the input into smaller units called tokens. These tokens are converted into numerical representations.

The attention mechanism then helps the model learn relationships between the tokens. For example, when reading a sentence, it can use the surrounding words to understand the meaning of a particular word.

The central attention equation is

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

Here, Q represents queries, K represents keys, and V represents values. These mathematical representations help the model decide which parts of the input deserve more attention.

Transformers also contain feed-forward layers that process the information learned by attention.

4.3 Where Are Transformers Used?

Transformers are used in a wide range of applications:

- Chatbots and AI assistants.
- Text generation and summarisation.
- Language translation.
- Question answering.
- Image recognition and processing.
- Systems that work with text, images, and audio.

4.4 Advantages and Limitations

Transformers can learn complex relationships in large datasets and process many parts of a sequence in parallel during training.

However, large Transformer models require significant computing power and memory. They can also produce incorrect information, even when their responses sound convincing.

5. Comparison of the Three Technologies

Although GANs, diffusion models, and Transformers are often discussed together, they solve problems in different ways.

Technology	Main Idea	Common Uses
GANs	Two networks compete to improve generated data.	Image generation and enhancement.
Diffusion Models	Noise is gradually removed to generate new samples.	Image, video, and audio generation.
Transformers	Attention learns relationships between input elements.	Language models, chatbots, and vision.

An important point is that GANs and diffusion models are generative approaches, whereas Transformers are a neural network architecture. A Transformer can also be used inside a diffusion model or another generative AI system.

6. Conclusion

GANs, diffusion models, and Transformers represent different approaches to building intelligent systems.

GANs learn through competition and can generate realistic images. Diffusion models learn to turn random noise into meaningful content. Transformers use attention to understand relationships in data and have become essential to modern language models.

Each technology has its own strengths and limitations. Understanding how they work provides a useful foundation for exploring generative AI and its growing applications in everyday life, education, research, and industry.